Alan Fong

DevOps & Automation Engineer

Johor Bahru, Malaysia · alanfong93@gmail.com
linkedin.com/in/alan-fong · github.com/alanfong93 · alanfong93.github.io



Professional profile

Hands-on IT manager combining infrastructure operations, software engineering, and AI-enabled automation. Established an SME's IT function as its first IT hire, grew the team, and continued to build and operate the systems supporting business operations. Experience spans technical support, team leadership, internal applications, distributed services, and workflow automation.

Professional experience

IT Manager · Modern Zone Marketing Sdn Bhd

Jun 2020 – Present · Johor Bahru, Malaysia

- Established the IT function as the first IT hire, covering infrastructure, networking, support, and software delivery for office and warehouse operations.
- Build and operate Hyper-V virtualization, virtual desktops, Docker services, reverse proxies, file storage, monitoring, and automated off-box NAS backups.
- Maintain the business network, including multi-WAN failover, DHCP, wireless access, and VPN access; deploy security keys and patch-management tooling.
- Deliver internal applications for asset tracking, attendance, orders, and support. Sole author of the C# asset-management system and rebuilt packaging recommendation service.
- Lead the Telegram AI-support bot and VM-management control plane with a collaborator, combining knowledge-base guidance, human escalation, job dispatch, and automated VM remediation.
- Build n8n workflows and scripts to automate recurring work, including sales/delivery order generation, reporting, and integrations between services.
- Grew the IT team from one to a peak of seven. Mentored a colleague with little programming background into the primary author of a shipped OCR automation.
- Introduced branch-based development, GitHub Issues, pull-request reviews, automated code review, and architecture/workflow/API documentation with Mermaid diagrams.

Chief / Senior Technician · Meadow Computer & System Sdn Bhd

2013 – Mar 2020 · Johor Bahru, Malaysia

Progressed from junior technician to senior and chief-technician responsibilities. Worked part-time during degree studies in 2014–2016 and returned after the 2018 software-engineering internship.

- Managed up to four technicians, including rosters, leave coordination, and training new technicians; ran after-sales and warranty services.
- Diagnosed computer issues, reinstalled operating systems, and installed routers and Wi-Fi mesh equipment.
- Digitized handwritten service/warranty forms using Google Forms and Sheets, and scripted recurring manual tasks.
- Maintained and extended an existing service/warranty management system, including WhatsApp customer messaging; maintained the software library used for customer installations.

Software Engineer Intern · GNey Software

2018 · Johor Bahru, Malaysia

Built project scaffolding, maintained client websites, and supported testing and documentation during the software-development internship.

Selected employer projects

Capabilities and contributions are described at a high level. Employer implementation remains private.

IT asset-management service

Contribution: sole author · C# / .NET 10, PostgreSQL

Clean Architecture/CQRS service for serialized asset assignments and consumable stock. EF Core persistence, permission-checked endpoints, ledger mutation enforcement, and bounded import work support the custody record.

MZM Project Board

Contribution: creator and primary author · Python, Docker, GitHub APIs

Module progress, blockers, and approval/review queues from read-only GitHub issue/PR data. Incremental API-budget-aware polling, persisted snapshots, refresh locking, and atomic publication protect refresh reliability.

Packaging recommendation service

Contribution: sole author of the rebuilt service and frontend · Python, FastAPI, JavaScript, OR-Tools

- Rebuilt an AI-generated prototype into a shipping-box recommendation service using OR-Tools CP-SAT for 3D packing with fragility, stacking, weight, and rotation constraints.
- Implemented background jobs with progress polling, recurring-scenario caching, and isolated worker processes to contain native solver failures.

Windows VM-management control plane

Contribution: led design and delivery with a collaborator · Python, PowerShell, Hyper-V

Central dashboard for authenticated job dispatch, batch PowerShell execution, and over-the-air agent updates across Windows VMs. A generic bundle system supports deployment of Windows Service executables.

Telegram AI-powered IT support

Contribution: led development with a collaborator · Python, Telegram, LLM integration

Checks staff requests against an FAQ and procedures knowledge base, guides users through known fixes, follows up on resolution, and escalates unresolved requests to an engineer. Includes common remediations such as VM restarts.

Distributed print-fleet service

Contribution: primary author of the printer service · Python, FastAPI, Raspberry Pi, Docker

Raspberry Pi agents expose USB printers through authenticated APIs with persistent, idempotent queues. Central enrollment, dispatch, callbacks, and over-the-air updates coordinate the fleet.

Scheduled broadcast automation

Contribution: primary author · Python, Pydantic, REST API

Scheduled audio playback with time- and season-aware playlists, a control/metrics API, circuit breakers, timeouts, and browser recovery. Validated configuration and hot reload support ongoing operation.

Warehouse, follow-up, and workflow automation

Contribution: primary author on warehouse work; contributor to follow-up tooling · JavaScript, n8n

- Maintained and extended warehouse tooling for multi-courier dispatch, consignment notes, and stock tracking within a multi-engineer codebase.
- Contributed to follow-up tooling; automated recurring reports and order-generation workflows.

Public personal projects

Network Sandbox · Live demo

Browser topology editor with scoped 802.1Q/802.1D bridge traces, routing/NAT/DHCP/ARP models, replay, and share-safe export. Not production certification or vendor emulation; radio coverage is not modelled.

ops-guard

Cited search, audited proposals, and catalog-bound MCP execution when an operator policy and execution catalog configure the gate. Server-side evidence/precondition/authorization checks; model risk assessments never authorize execution.

local-judge

Typed judgments through Python, HTTP, and MCP with code-side validation and a documented Jev-compatible subset. Ollama/OpenAI-compatible adapters; sample agreement is not calibrated confidence. Representative live-model usefulness remains unproven.

jiandu

Source-backed context kit preserving originals with cited wiki notes, compile/index skills, and MemPalace pointers for Markdown and captions.

Additional personal engineering

Mobility World

r5py/R5 before/after comparisons over pinned OSM/GTFS data, synthetic routes, and an assumption-aware line editor. Three.js/Rapier walk/drive/motorcycle prototype. Cost ranges and input provenance; not observed traffic or calibrated congestion. Hardware-performance acceptance remains outstanding.

JoJo: OpenCode/n8n assistant tooling plus a Microsoft Agent Framework specialist runtime with typed dispatch, durable admission/checkpoints, bound human approvals, effect journals, and operator controls. MemPalace/Obsidian provide context.

AI Eyes: FastAPI camera-to-AI prototype for identification, how-to guidance, and translation.

Other tooling: WhatsApp/Telegram messaging APIs, schema-aware database tools, Windows scripts, and reusable agent skills.

Ad-hoc freelance IT support

Selected paid engagements: 2018–2022 and 2025

Provided IT troubleshooting for individual clients. Later remote engagements included application installation and software-licence troubleshooting.

Technical skills

Engineering: Python · C# / .NET · PHP · JavaScript · TypeScript · PowerShell · Bash · FastAPI · EF Core / PostgreSQL

Infrastructure and operations: Docker / Compose · Hyper-V · Proxmox VE · Linux · Nginx · AWS · Google Cloud Platform · Google Workspace · Uptime Kuma · Synology storage and backups

Networking and security: OpenWRT · pfSense · multi-WAN · VPN · Cloudflare Tunnel · YubiKey · Action1 patch management

Automation and AI: n8n · MCP · OpenCode · Microsoft Agent Framework · Ollama · Whisper · Telegram Bot API · AI-assisted development and review

Education and certification

Bachelor of Software Engineering (Hons) · Southern University College
2014–2018 · Second Class Upper Division · Conferred 27 May 2018

Google IT Support Professional Certificate · 2024